

# The Easy Twenty

**India can stretch ethanol blending to E25–E27 this decade. The road beyond runs through flex-fuel vehicles it barely sells, a fuel value chain that barely exists, and a feedstock base that rises and falls with the monsoon — and the crude-import prize, measured honestly, is smaller than the ambition.**

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## Executive Summary

India crossed 20% ethanol blending in late 2025 and made E20 its standard petrol grade from April 2026 — a national target met six years early. The question now facing investors is whether the programme can climb higher: to E25 or E30 by 2030, to flex-fuel and E85 beyond that, and what any of it actually does to India's crude import bill. We set out our answer in this report.

Our answer is that India has just finished the easy part. The climb from 1.5% to 20% blending was achieved by pouring a little ethanol into engines the country already owned, on a feedstock base it already grew, with no change asked of the consumer. None of those three conditions holds for the next leg. The honest verdict, which this report defends in detail, is that E25 to E27 is reachable by 2030 and worth banking — but it is the ceiling of the cheap, easy programme, not a waypoint on the road to E85. Everything past it is a different and much harder project.

Start with the physics, because it sets the boundary. Ethanol carries about 65% of petrol's energy by volume, and the vehicles already on Indian roads were designed for petrol. There is a practical limit — the blend wall — beyond which you cannot push ethanol into an ordinary engine without mileage loss, cold-start trouble, and corrosion. For India's legacy fleet that wall sits at about E20. New engines tuned for E20, mandatory since April 2025, give a little headroom, which is why E25 and even E27 are achievable as the fleet turns over. But that headroom is measured in single digits, not multiples. The world confirms the boundary: the United States, the largest ethanol producer on earth, has been stuck at E10 for fifteen years; Europe caps food-crop biofuel at 7% and bets on electrification. Nobody blends their way past the wall. They either stop, or they change the car.

Then there is the feedstock — and this is the part of the story that gets least attention and

deserves most. India's ethanol does not come from a stable industrial input. It comes from the fields, and the fields depend on the rain. The programme learned this the hard way. When a poor monsoon in 2023 cut the sugar crop, the government banned sugarcane juice for ethanol to protect the sugar bowl, and distilleries pivoted to maize. The consequences were immediate and instructive: maize demand exploded, prices jumped by half, and India — a country that had exported two to four million tonnes of maize a year — became a net importer for the first time in a generation, buying corn from Myanmar and Ukraine. Pause on what that means. The entire point of ethanol is to replace an import, crude oil, with something grown at home. For two years the programme was substituting one import bill for another, paying more for the costliest feedstock in the system. In September 2025, with the monsoon recovered and sugar back in surplus, the government reversed the ban and cane flooded back. The lesson is not that the system is broken. It is that the feedstock base swings with the weather, and the higher you set the blend, the more a single bad monsoon threatens the whole structure. Sugarcane, the cleanest feedstock, is also among the thirstiest crops India grows, concentrated in its most water-stressed states. Building a high-blend future on it is building on a floodplain.

This is where Brazil — the one country that genuinely blends high — both tempts and misleads. Brazil runs E30 today, with ninety per cent of its cars able to burn any blend up to pure ethanol, and a pricing system that lets the blend and the pump price float with the sugar-versus-ethanol economics each year. It is elegant, and it is the right model in principle. But Brazil built it over fifty years on abundant land, abundant water, world-beating cane yields, and — crucially — consumer choice at the pump. India administers its ethanol price by fiat, gives the driver no choice of blend, and sells E20 at the same price as petrol despite the lower energy. You cannot run Brazil's dynamic market on India's command-and-control plumbing without rebuilding the plumbing first.

Which brings us to the only real road beyond the blend wall: flex-fuel vehicles. The launches have arrived — Maruti's WagonR Bioflex and Hero's flex Splendor and HF Deluxe, all in June 2026, alongside the first E85 pumps. This is a genuine milestone, and we treat it as one. But a value chain is three things — vehicles, fuel, and the feedstock behind the fuel — and India has the beginnings of one and barely any of the other two. E85 sells at 48 outlets nationwide against a target of five thousand. And the economics, today, do not work for the buyer: E85 is priced about 20% below E20, but a flex vehicle burns roughly a third more of it, so running on E85 currently costs more per kilometre, not less. Industry's own people say the discount needs to reach 30% before the maths turns. Until it does, the flex car is a compliance product, not a consumer choice. We see meaningful flex penetration as a 2035-to-2040 story, contingent on a fuel network, a deeper discount, and tax parity that does not yet exist.

And then the question that should anchor every investor's mind: even if it all works, what is the prize? Here the arithmetic is sobering. Petrol is only about a sixth of India's oil

consumption — the other 91 million tonnes is diesel, which ethanol cannot touch. Passenger vehicles are only about a third of petrol. So if you did the hard thing — converted the entire passenger-vehicle fleet to flex and ran it on E85 — you would displace, at the theoretical maximum, something in the order of 4-5% of the crude import bill, and you would need more ethanol than India's entire installed distillery capacity to do it. A realistic 2035 penetration of a quarter of the PV fleet saves closer to one per cent. The passenger-car flex-fuel story, in other words, is real, slow, and — for the crude bill — small. The larger petrol prize sits in two-wheelers, which are 61% of petrol demand; but those are also the vehicles fleeing fastest to electric. Ethanol's window in the segment that matters most is the one EVs are closing.

The investment posture that falls out of this is not bearish on ethanol. It is precise about where the value is. E25–E27 by 2030 is the bankable win, and it rewards integrated, feedstock-flexible producers who can swing between cane and grain as the monsoon dictates. Flex-fuel is a long-dated option, not a near-term volume story, and it lives or dies on policy levers — the E85 discount, GST parity, the retail build-out — not on consumer economics as they stand. And the auto winners through this decade are the makers of E20-calibrated engines and strong hybrids, not the flex-fuel pioneers. The ceiling is real. India just hasn't hit the parts of it that hurt yet.

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## Investment Conclusion

Our central view is that India's ethanol programme has a hard, near-term ceiling that the market is not yet pricing. E25 to E27 nationwide blending is achievable by 2030 — the installed distillery capacity exists, the cane reinstatement has restored feedstock flexibility, and the engine fleet is turning over to E20-tuned units that can tolerate the stretch. We would treat that as the bankable base case. Beyond E30, blending into the conventional fleet is not feasible; the blend wall is a physical limit, not a policy choice, and no major economy except Brazil has crossed it. Progression past E30 is therefore not a fuel-mandate question at all. It is a vehicle question, and it depends entirely on flex-fuel adoption.

On flex-fuel, we are constructive on the direction and cautious on the pace. The June 2026 launches from Maruti and Hero are real and matter as proof that the vehicles can be built and homologated at India's price points. But the value chain behind them — a national E85 retail network, a dedicated high-blend feedstock supply, and consumer economics that reward the switch — is embryonic. At today's roughly 20% E85 discount the running-cost case is negative for the buyer once the energy penalty is accounted for; the discount needs to reach about 30% to turn, and that gap is a fiscal subsidy the exchequer has not committed to permanently. Flex-fuel vehicles also carry an 18–40% GST burden against 5% for EVs. We see a meaningful flex parc — a quarter or more of passenger vehicles — as a

2035–2040 proposition, not a this-decade one.

The prize, even on success, is bounded by structure. Ethanol addresses only the petrol pool, roughly a sixth of India's oil consumption; the 91-million-tonne diesel pool is untouched. Within petrol, passenger vehicles are about a third. On our arithmetic, full conversion of the PV fleet to E85 would displace on the order of 4–5% of the crude import bill at the theoretical maximum — and would require more ethanol than India currently has capacity to make. A realistic mid-2030s penetration saves closer to 1%. We would therefore underwrite this theme as an agricultural-margins and energy-security story with a real but capped endpoint, not as a structural answer to India's oil dependence. The bankable trade is feedstock-flexible producers riding the E25–E27 stretch and the E20-engine and strong-hybrid makers; flex-fuel is an option we would buy cheaply and hold patiently, with the E85 discount, GST treatment, the monsoon, and the EV curve as the variables that decide it. The single most important risk to our caution is crude itself: sustained oil above \$90–100 per barrel rewrites the economics and could pull the whole timeline forward.

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## **1. India Has Finished the Easy Climb — and the Road Now Forks Into Two Hard Ones**

The first thing to understand about India's ethanol programme is why it succeeded so quickly, because the reasons it succeeded are precisely the reasons it now slows. Blending rose from under 2% a decade ago to 20% in 2025, ahead of every official timeline. That was not a triumph of technology or a feat of industrial construction. It was the result of three favourable conditions, each of which is about to disappear.

The blend was poured into engines India already owned. Adding 10%, then 20%, ethanol to petrol required no new vehicle, only minor changes to rubber and seals on new models. The feedstock came from crops India already grew, principally sugar by-products that mills were glad to monetise. And the consumer was asked for nothing — no new pump, no price decision, no behaviour change. Ethanol simply appeared in the tank. A programme that asks nothing of anyone is easy to scale. That is the story of the first twenty per cent.

Every one of those conditions reverses now. Past the blend wall, ethanol needs a different engine — a flex-fuel vehicle the consumer must choose to buy. Past the easy feedstock, ethanol needs either more cane from a water-stressed land or more grain from a food supply, both of which create new tensions. And past E20, somebody has to pay: the driver, in a vehicle premium and a fuel choice, or the exchequer, in a subsidy. The next leg of this programme asks something of everyone.

So the road forks, and both forks are steep. One fork is to push a higher blend — E25, E30 — into the existing fleet. The other is to put a new kind of vehicle, the flex-fuel car, into the

parc and feed it E85 or E100. This report walks both forks, because the government is signalling both at once, and they lead to very different places. The first fork dead-ends at the blend wall within a few percentage points. The second fork is open-ended but barely paved. Which fork India can actually travel, how far, and to what benefit, is the whole question.

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## **2. Blended Into the Engines India Already Owns, Ethanol Stops at the Blend Wall — and That Wall Is About E25**

Take the first fork first, because it is the one the market assumes is open. The assumption is wrong past a point, and the point is closer than the roadmaps suggest.

There is a physical ceiling to how much ethanol an ordinary petrol engine can burn, and it is not a matter of will. Ethanol holds roughly 65% of petrol's energy by volume, absorbs water, and corrodes materials that petrol leaves alone. At low blends these effects are trivial. As the blend rises they compound — mileage falls, cold starts get harder, fuel-system components degrade in vehicles never designed for them. The blend at which these problems become unacceptable for the legacy fleet is, by broad international consensus and India's own engineering experience, about 20%. This is the blend wall. It is why E20, not some higher number, is where India planted its flag.

India does have a little room above the wall, and it is worth being precise about how much. Since April 2025, new vehicles must be built with E20-tuned engines, and these can exploit ethanol's high octane through higher compression to claw back some of the energy penalty. As the fleet turns over to these engines through the late 2020s, the tolerable blend creeps up. That is the real, defensible case for E25 and even E27 by 2030: not a breakthrough, but a slow ratchet as old engines retire and new ones cope with a little more ethanol. The regulatory scaffolding is already in place — specifications for E22 through E30 are notified, and the excise duty on those grades has been waived. E25 by 2030 is a credible base case.

But notice what this is, and what it is not. It is a stretch of single percentage points, bought by replacing the entire vehicle fleet over years. It is not a ramp to E50 or E85. You cannot reach those numbers by blending into conventional engines, however new, because the wall does not move that far. The clinching evidence sits abroad. The United States produces more ethanol than any nation in history and has mandated it aggressively for two decades; it remains stuck at E10, because pushing higher blends into ordinary cars does not work and the economics of forcing it are punishing. Europe caps food-crop biofuel at 7% by deliberate choice and routes its decarbonisation through electrification. Neither is short of ambition or ethanol. Both have found the same wall.

The conclusion from the first fork is therefore clean, and it bounds everything that follows.

Blending into the fleet India owns and is buying can reach E25–E27 this decade. It cannot, on any realistic engineering, go materially beyond E30. Anyone who tells you India will blend its way to E50 is describing a fuel no ordinary engine can use. Past E30 is not a blending story. It is a vehicle story — the second fork — and we come to it after dealing with the constraint that limits even the first.

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### **3. The Plants Are Built; the Feedstock Is Rain-Fed — and That Is the Real Constraint**

Even the modest stretch to E25–E27 runs into a constraint that the capacity figures hide. India does not lack distilleries. It lacks reliable feedstock — and the reason is that its feedstock grows in fields that depend on the monsoon.

Begin with what is not the problem. India has built roughly 18–20 billion litres of installed ethanol capacity, against E20 demand of about 11 billion litres. The plants run at barely 25–30% utilisation. On paper there is room for E30 and beyond several times over, which is exactly why the distilling lobby pushes so hard for higher blends — they have built capacity that has nothing to do, and a higher mandate fills it. Capacity, in short, is not the binding constraint. It is a glut. As Exhibit 1 would show — installed capacity against ethanol actually required at E20, E25 and E30, with the gap shaded — the plants can supply E30 today. The question is whether the fields can.

They can, in a good year. The deeper limit is sustainable feedstock — how much cane and grain India can divert to fuel without starving its sugar bowl or its food plate or its water table. Independent estimates put that sustainable ceiling at roughly 6–10 billion litres; the government's own optimistic figure is about 13.5 billion. The gap between those numbers is the gap between "E20 is already the limit" and "E25 has headroom." And the reason the estimate is a range rather than a number is that the answer changes every year with the rain.

Here is the episode that proves it, and it is the most important passage in this report. In 2023 a weak, El Niño monsoon — India's poorest rainfall in five years — hit the cane states hardest; Maharashtra and Karnataka, nearly half the national crop, saw yields collapse from 120–130 tonnes a hectare toward 80. With sugar suddenly scarce, the government did the politically obvious thing: in December 2023 it banned the diversion of sugarcane juice to ethanol, to keep sugar on shelves. Distilleries, needing a feedstock, turned to maize. And the consequences ran exactly counter to the programme's purpose. Maize demand surged; prices climbed by roughly half, to ₹24,000–25,000 a tonne; the poultry and feed industries, which rely on maize, were outbid by fuel. India, long a maize exporter of two to four million tonnes a year, became a net importer for the first time in a generation — buying nearly a

million tonnes from Myanmar and Ukraine in 2024, a figure up almost eighty-fold on the year before, while its own exports collapsed. Maize-based ethanol became the single costliest feedstock in the system at ₹71.86 a litre, after the government raised its price 29% to attract supply.

Sit with what that means, because it punctures the headline rationale. Ethanol exists to swap an import — crude oil — for something grown at home. For two years, to make that ethanol, India was importing maize. It had traded an oil import for a grain import, and paying a premium for the privilege. The forex saving the programme is built to deliver was leaking out the back door. This was not a design feature. It was what happened when the monsoon failed and the feedstock base buckled.

In September 2025 the government reversed the ban. The monsoon had recovered, two good years had refilled the reservoirs and the cane fields, sugar was back in surplus, and cane juice and molasses flooded back into the ethanol stream. Sugar-company shares jumped. The food-versus-fuel tension eased as fast as it had arrived. The system, in other words, is not permanently grain-led or permanently cane-led. It is monsoon-led. It tilts to cane when the rain is good and the sugar surplus large, and lurches to grain — with all the import leakage that entails — when the rain fails. That swing is the structural fact. As Exhibit 2 would show — feedstock mix by year against monsoon rainfall, with the 2023 ban and 2025 reversal marked — the feedstock follows the weather, not a plan.

This is why the cane reinstatement, welcome as it is, does not resolve the constraint so much as relocate it. Cane is the cleanest feedstock per litre of ethanol, but it is among the thirstiest crops India cultivates, and it grows in precisely the states — Maharashtra, Karnataka — where water stress is most acute. Leaning the ethanol programme back onto cane eases the food fight and reopens the water one. And it leaves the whole edifice exposed to the next poor monsoon, which will come. The higher the blend India targets, the more ethanol it must draw from these fields, and the more a single dry year threatens not just sugar prices but the fuel supply itself. You can stretch to E25 on this base in a good year. Betting E30 on it every year is betting on the rain.

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## **4. Brazil Lets the Market Price Ethanol; India Prices It by Fiat — Which Is Why India Cannot Flex the Way Brazil Does**

The natural reply to all this is: Brazil manages. Brazil runs E30, sells pure ethanol at every pump, and absorbs bad cane years without crisis. Why can India not simply copy the model that visibly works? The question is the right one, and the answer reveals why the second fork is so much harder than it looks.

Brazil's genius is not its blend level. It is its pricing. Brazil does not fix one blend and one

price and hold them against the weather. It lets both float. The anhydrous blend in ordinary petrol is set in a band — historically 18 to 27%, now 30% — adjusted with the relative price of sugar and ethanol. More importantly, ninety per cent of Brazil's cars are flex-fuel, and at the pump the driver chooses between gasoline and pure hydrous ethanol on a simple rule of thumb: ethanol pays when it is priced below about 70% of gasoline, reflecting its lower energy. When cane is short and ethanol dear, drivers buy gasoline and ethanol demand falls automatically. When cane is plentiful and ethanol cheap, they switch back. The market clears itself. A bad monsoon — or its Brazilian equivalent — raises the price, dampens demand, and the system breathes. Price is the shock absorber.

India has built the opposite machine. Its ethanol price is administered — set by the government, feedstock by feedstock, delinked from crude, revised by decree. Its driver has no choice of blend; E20 is what comes out of the pump, and increasingly E0 is not available at all. And the price the driver pays does not fall to reflect ethanol's lower energy — E20 sells at the same rupee-per-litre as the petrol it replaced, even though it carries less energy, so the consumer quietly absorbs the penalty. There is no shock absorber here. When the feedstock base buckles, as it did in 2023, the system cannot price its way to balance; it can only ration by decree — ban cane, divert maize — with the second-order consequences we have just traced.

This matters enormously for the high-blend ambition, because flex-fuel only delivers its promise with flexible pricing behind it. The whole point of a flex car is that the driver switches fuels as the economics shift. Strip out the price signal — give the consumer a single administered blend at a price that does not move with feedstock economics — and you have the hardware of the Brazilian system with none of its intelligence. Adopting Brazil's model is not a matter of mandating flex vehicles. It would require dismantling India's administered-price architecture, building genuine blend choice at the pump, and letting the ethanol price float — a structural reform of the fuel market that the government has shown no sign of contemplating, because it would mean surrendering a lever it values. Until that changes, India can buy Brazil's cars. It cannot run Brazil's market.

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## **5. Beyond the Blend Wall Lies a Flex-Fuel Value Chain That, Today, Barely Exists**

Now the second fork, in full. If India wants to go past E30, it must put flex-fuel vehicles into the parc and feed them high blends. In June 2026 it took the first real step: Maruti launched the WagonR Bioflex, India's first flex-fuel passenger car, able to run anything from E20 to pure ethanol; Hero launched flex versions of the Splendor and HF Deluxe, its volume motorcycles; and the first E85 pumps were inaugurated in Delhi and a handful of cities. We do not dismiss this. It is the genuine beginning of the only path that leads anywhere beyond



the blend wall. But a beginning is what it is.

A flex-fuel value chain has three links, and a chain is only as strong as its weakest. The first link is the vehicle, and here India has made real progress — Maruti and Hero have shown the cars and bikes can be built and certified at Indian price points, and others have prototypes. The second link is the fuel: a national network of E85 pumps dense enough that a flex owner is never stranded. Here India has 48 outlets against a stated target of 5,000 by end-2027 — which is to say, it has a pilot, not a network. The third link is the feedstock: a dedicated, reliable supply of ethanol large enough to feed a high-blend fleet, which runs straight back into the rain-fed constraint of Chapter 3. India has the first link in prototype, the second in pilot, and the third hostage to the monsoon. That is not yet a value chain. It is the sketch of one.

And the link that will decide adoption is the one the launches cannot fix by themselves: the economics for the buyer. A flex car costs more — the WagonR Bioflex carries roughly an ₹86,000 premium over its conventional sibling — and that premium is meant to be repaid through cheaper fuel. But cheaper per litre is not cheaper per kilometre. E85 launched at about ₹82 against E20 at about ₹102 — a 20% discount that looks attractive until you remember the energy. E85 carries roughly 30% less energy than E20, so a flex vehicle burns about a third more of it to cover the same distance. Do the arithmetic and, at a 20% per-litre discount, running on E85 costs more per kilometre than running on E20, not less. The premium never repays. Industry's own engineers concede the threshold: the discount has to reach about 30% before the per-kilometre maths turns positive. The government is delivering 20%. The single most important number for flex-fuel adoption is pointed the wrong way.

There is also the matter of how the state taxes the choice. Flex-fuel vehicles sit in the ordinary 18–40% GST brackets; battery EVs sit at 5%. A government that genuinely wanted flex-fuel to win would not handicap it sevenfold against the electric alternative it is supposedly an alternative to. The tax signal says the opposite of the launch-day rhetoric.

The world offers a sobering precedent for exactly this moment. Thailand did what India is now attempting — built flex vehicles, subsidised E85 to an attractive discount, stood up the pumps. It worked, for a while. Then the subsidy bill became unbearable, demand for E85 stayed thin, EVs began eroding the fleet, and in early 2026 the state-owned retailers simply stopped selling E85, with a full phase-out scheduled and the surplus ethanol redirected to aviation fuel. China went further back and gave up earlier, abandoning a national E10 target it never reached. The pattern is consistent: high-blend programmes that depend on a permanent subsidy to make the consumer maths work tend, eventually, to lose the argument with the finance ministry. India's flex-fuel future rests on a discount it must fund forever, against an EV alternative it taxes far more lightly. We see meaningful flex penetration — a quarter or more of the fleet — as a 2035–2040 possibility at the earliest,

and only if the discount deepens, the network is built, and the tax treatment is fixed. None of those three is yet in hand.

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## **6. The Prize, Measured: Even a Fully Flex Passenger-Vehicle Fleet Barely Dents the Crude Bill**

Suppose, though, that it all works. Suppose the network gets built, the discount deepens, the tax is fixed, and India converts its passenger-vehicle fleet to flex-fuel running on E85. What does the country actually save on its crude bill? This is the question every investor should ask before underwriting the theme, and the answer is the quiet heart of this report. The prize is real. It is also small.

The arithmetic is bounded before it begins by two structural facts. First, ethanol is a petrol substitute, and petrol is only a sixth of India's oil. Of roughly 239 million tonnes of petroleum products India consumes, about 91 million tonnes is diesel — trucks, buses, tractors, rail — which ethanol cannot touch at all, and only about 40 million tonnes is petrol. Second, within that petrol pool, passenger vehicles are only about a third; two-wheelers are the larger share at 61%. So when the user asks specifically what passenger vehicles can save, the question is already scoped to roughly a third of a sixth — about 18 billion litres of petrol a year out of an oil system many times larger.

Now run the conversion at its theoretical maximum. Put the entire passenger-vehicle fleet on E85. Because E85 is mostly ethanol and carries less energy, replacing the energy in that 18 billion litres of petrol displaces — after accounting for the petrol still in the blend and the extra volume burned — on the order of 14 billion litres of petrol, or roughly 10 million tonnes. Set against crude imports of about 234 million tonnes, that is in the order of 4-5% of the import bill. And it is not free: it would require well over 20 billion litres of ethanol a year — more than India's entire installed distillery capacity today — drawn from the rain-fed feedstock base of Chapter 3, to be grown every year regardless of the monsoon. As Exhibit 3 would show — crude displaced as a share of imports under 25%, 50% and 100% PV flex conversion, against the ethanol required for each — the maximum theoretical saving sits in the low single digits of the import bill, at an ethanol cost the country cannot currently supply.

The realistic figure is smaller still. A flex fleet does not appear by decree; it accretes as people buy flex cars over fifteen years of fleet turnover. A plausible penetration by the mid-2030s — a quarter of the passenger-vehicle parc on E85 — displaces perhaps 3-4 billion litres of petrol, around 2-3 million tonnes, or roughly 1% of the crude bill. One per cent, for the successful execution of a flex-fuel value chain that does not yet exist, against a feedstock base that depends on the rain. That is the honest measure of the passenger-

vehicle prize.

The point is not that ethanol is worthless — at E20, blended across all petrol vehicles, it already displaces a more meaningful slice, and the farm-income and energy-security arguments have weight. The point is that the marginal high-blend ambition, the flex-fuel leap that the launches herald, buys remarkably little crude relief for the difficulty involved, when measured against the whole oil bill. The larger petrol prize lies not in passenger cars but in two-wheelers, that 61% of petrol demand. But two-wheelers are also the segment electrifying fastest, where EVs already win on running cost in cities and the government's own tax code favours them. Ethanol's biggest opportunity and its strongest competitor occupy the same vehicles. The window is closing from the other side.

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## **7. Conclusion: Bank the Twenty-Five, Doubt the Eighty-Five**

India's ethanol programme deserves the credit it has earned. Reaching E20 across a vast and chaotic vehicle fleet, six years early, on a domestic feedstock, is a genuine achievement of policy and logistics, and the farm-income and energy-security benefits are not imaginary. We begin from respect for what has been done.

But the questions this report set out to answer have clear answers, and they are more sober than the launch-day mood. Can India reach E25 or E30 by 2030? E25 to E27, yes — it is a bankable base case, achieved by the slow turnover of the fleet to E20-tuned engines and supplied, in good monsoon years, by a cane sector restored to flexibility. E30 is the frontier of what blending into conventional engines can do, and it is monsoon-dependent. Can it go beyond? Not by blending — the wall is physical, and only Brazil, with endowments and a pricing market India does not have, has crossed it. Beyond E30 requires flex-fuel vehicles and a value chain — cars, pumps, dedicated feedstock — that India has only just begun to assemble, whose consumer economics are currently negative, and which we judge a 2035–2040 proposition at best. And the prize, isolated to passenger vehicles as asked, is small: of the order of 1% of the crude bill at realistic penetration, 4–5% at a theoretical maximum that would consume more ethanol than India can make. The diesel pool, two-thirds of the oil problem, is untouched throughout.

The frame that holds all of this together is the monsoon. India has built an energy ambition on an agricultural base, and that base rises and falls with the rain. As long as the ambition stays modest — E20, blended invisibly into existing engines — a rain-fed foundation holds, because a bad year can be absorbed by a temporary policy patch, as 2023 showed. The higher the ambition climbs — E30, flex fleets, billions in distillery and retail capital, a fuel network the country depends on — the more a single dry monsoon threatens the whole structure, and the more the programme's forex logic leaks when it is forced onto imported grain. That is the ceiling. It is not a number on a chart. It is the weather.

For investors the posture is therefore neither bullish nor bearish but located. The value through this decade is in the E25–E27 stretch and the producers flexible enough to feed it from whichever crop the monsoon favours, and in the makers of E20 engines and strong hybrids who win the fleet regardless of how the flex story plays. Flex-fuel is an option to hold patiently and price cheaply, decided by levers — the E85 discount, GST parity, the retail roll-out — rather than by economics as they stand. We would change this view on three signals: a sustained crude price above \$90–100 that rewrites the maths, a firm and funded E85 discount near 30% with tax parity to match, or a genuine breakthrough in second-generation ethanol that finally cuts the feedstock free from the fields and the rain. Absent those, the conclusion stands. India has blended the easy twenty. The hard part starts now, and it is harder than the ambition admits.

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*Data notes: petrol consumption ~40 MMT and total petroleum products ~239 MMT (PPAC, FY2024–25); diesel ~91 MMT; crude imports ~234 MMT at ~88% import dependence. Petrol segment split (two-wheelers ~61%, passenger vehicles ~34%, three-wheelers ~2–3%) per PPAC/Nielsen study. Ethanol carries ~65% of petrol's volumetric energy; E85 ~70% of E20's. Installed ethanol capacity ~18–20 billion litres at ~25–30% utilisation; E20 demand ~11 billion litres; sustainable feedstock ceiling ~6–10 billion litres (independent estimates) versus ~13.5 billion litres (NITI Aayog 2021). Sugarcane concentration in Maharashtra, Karnataka and UP (>80% of output) and 2023 El Niño yield collapse per ISMA/USDA FAS and trade reporting; December 2023 cane-juice diversion ban and September 2025 reinstatement per government notifications. Maize: India a net importer in 2024 (~0.9 MT, up ~7,940% YoY; exports down from 2–4 MT historically) per Ministry of Commerce/USDA FAS; maize-ethanol procurement ₹71.86/L after a 29% increase. Maruti WagonR Bioflex (~₹7.24 lakh, ~₹86,000 premium) and Hero flex Splendor+ (₹82,710)/HF Deluxe (₹72,792) launches June 2026; E85 ~₹82.12 vs E20 ~₹102.12 at launch across 48 outlets, target 5,000 by end-2027. Brazil E30 from August 2025, ~90% flex fleet, ~70% ethanol/gasoline pump-parity rule, sugar-ethanol-linked blend band, per CNPE/USDA FAS/IFPRI. All ethanol-requirement and crude-displacement figures are analyst estimates derived from first principles on the stated petrol base and are illustrative, not forecasts; government benefit claims (forex saved, CO2 avoided, farmer income) are self-stated and not independently audited. Crude-price sensitivity, GST rates (flex-fuel 18–40% vs EV 5%) and EV-penetration trajectories as cited in text.*